National University of Computer and Emerging Sciences

Lahore Campus

GenAI (AI4009)

Date: April 12th 2025

Course Instructor:

Dr. Hajra Waheed

Sessional-II Exam

Total Time (Hr): 1

Total Marks: 20

Total Questions: 2

[redacted] Roll No

[redacted] Section

[redacted] Student Name

Attempt all questions in the provided sequence on your answer sheets. Marks to be deducted otherwise.

CLO 2: Analyze the architectures and pre-training methodologies of large language models.

CLO 4: Discuss advanced topics such as prompt engineering, retrieval-augmented generation (RAG), fine-tuning techniques, quantization.

Q1: Answer the following short questions. **[3 + 3.5 + 4 marks]**

a) Mistral 7B has 32 layers. If each token stores 16 bytes of key-value (KV) cache per layer, how much memory (in MB) is needed to store the KV cache for a batch of 8 sequences, each with 512 tokens?

b) How is prefix tuning different from prompt tuning? Explain.

c) What is the QLoRA's double quantization phenomenon and how is it beneficial?

CLO 2: Analyze the architectures and pre-training methodologies of large language models.

Q2: You are given a batch of 2 sequences, each of 22 tokens.

Each token has a 6-dimensional embedding.

You process them using chunked attention with the following setup:

- Chunk size: 4 tokens per chunk
- For each sequence, chunks are processed independently

For all chunks:

- Q is computed individually for each chunk
- K_all, V_all are accumulated across chunks within a sequence
- If a chunk has fewer than 4 tokens (e.g., remainder), process it as-is (no padding)

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Considering the above scenario, **answer the following questions.** **[2 + 3.5 + 2 + 2 marks]**

a) How many chunks will be created per sequence?

b) For Chunk 3 of Sequence 1, what are the dimensions of:

- Q 4x1
- K 4x1
- V 4x6
- K_all 12
- V_all 12

c) Suppose now that chunk 4 only takes the context from the immediate previous chunk for K_all, V_all. What would their dimensions be in Chunk 4? 16x1

d) What would happen if we do not store K_all and V_all across chunks? What impact it has?

$$\begin{bmatrix} T_{11} & T_{12} & T_{13} & T_{14} & T_{15} & T_{16} \\ T_{21} & T_{22} & T_{23} & T_{24} & T_{25} & T_{26} \\ T_{31} & T_{32} & T_{33} & T_{34} & T_{35} & T_{36} \\ T_{41} & T_{42} & T_{43} & T_{44} & T_{45} & T_{46} \end{bmatrix} \begin{bmatrix} - \\ - \\ - \\ - \end{bmatrix} =$$

~~6x6~~ $\boxed{4 \times 6}$ 6x1